

## SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – FEB / MARCH 2025

|             |                                                                |            |                |
|-------------|----------------------------------------------------------------|------------|----------------|
| Program     | : <b>B.E :- Common to all Programs</b>                         | Semester   | : <b>V</b>     |
| Course Name | : <b>Research Methodology and Intellectual Property Rights</b> | Max. Marks | : <b>100</b>   |
| Course Code | : <b>AL58 (2021,2022 Batch)</b>                                | Duration   | : <b>3 Hrs</b> |

### Instructions to the Candidates:

- Answer one full question from each unit.
- Use of Statistical tables is permitted.

### UNIT - I

- With relevant flow chart describe the necessary steps involved in research process. CO1 (06)
  - Distinguish between descriptive and analytical research with example. CO1 (06)
  - What are the key features of the bibliographic database of the Web of Science (WoS), and how is it commonly used in research? CO1 (08)
- How does the new and existing knowledge can contribute to the research process? Explain with relevant points. CO1 (06)
  - What are the primary goals of conducting a literature review in academic research? CO1 (06)
  - Write a note on different types of research misconduct. CO1 (08)

### UNIT - II

- What is a Research Design? What is the need for Research Design? CO2 (10)
  - What are experimental and control groups? Why are they important in research? How do they impact the research outcomes? CO2 (10)
- What is a randomized block design? what are its advantages and applications? CO2 (08)
  - State the principles of experimental design. CO2 (05)
  - How do you apply Latin square design for experimentation? Explain. CO2 (07)

### UNIT - III

- Discuss different statistical measurement scales. CO3 (10)
  - Suppose we are interested in a population of 20 industrial units of the same size, all of which are experiencing excessive labour turnover problems. The past records show that the mean of the distribution of annual turnover is 320 employees, with a standard deviation of 75 employees. A sample of 5 of these industrial units is taken at random which gives a mean of annual turnover as 300 employees. Is the sample mean consistent with the population mean? Test at 5% level. CO3 (10)
- Explain the procedure for hypothesis testing. CO3 (10)
  - The null hypothesis is that 20 per cent of the passengers go in first class, but management recognizes the possibility that this percentage could be more or less. A random sample of 400 passengers includes 70 passengers holding first class tickets. Can the null hypothesis be rejected at 10 per cent level of significance? CO3 (10)

## UNIT- IV

- |    |    |                                                                                  |     |      |
|----|----|----------------------------------------------------------------------------------|-----|------|
| 7. | a) | Discuss the different types of patent applications in India.                     | CO4 | (10) |
|    | b) | Describe the various grounds for Post-grant opposition as applied to Patents.    | CO4 | (10) |
| 8. | a) | Describe briefly the various forms of Intellectual Property Rights with example. | CO4 | (10) |
|    | b) | Explain the principles underlying patent law in India.                           | CO4 | (10) |

## UNIT - V

- |     |    |                                                                                                                                        |     |      |
|-----|----|----------------------------------------------------------------------------------------------------------------------------------------|-----|------|
| 9.  | a) | What are the rights conferred by registration of the copyright? Explain.                                                               | CO5 | (10) |
|     | b) | Explain different categories of trademarks recognized under Indian law, and tabulate the famous trademark types with examples.         | CO5 | (10) |
| 10. | a) | What are Trademarks? Why should we register the trademark? What can be registered as Trademark?                                        | CO5 | (10) |
|     | b) | Explain the following with respect to Copyright. (i) Moral and Economic Rights (ii) Duration of protection (iii) Author and Ownership. | CO5 | (10) |

\*\*\*\*\*